Overview of ML and AI

What Is Machine Learning?

- D1: The science of programming computers so they can learn from data

- D2: The field of study that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959)

- D3: A computer program is said to learn from experience E with respect to some task T and some performance metric P, if its performance on T, as measured by P, improves with experience E (Tom Mitchell, 1997)

What Is Machine Learning?

- Spam filter
  - T is flagging spam for new emails
  - E is training dataset
  - P is ratio of correctly classified emails (accuracy)
  - Spam filter is an example of classification task

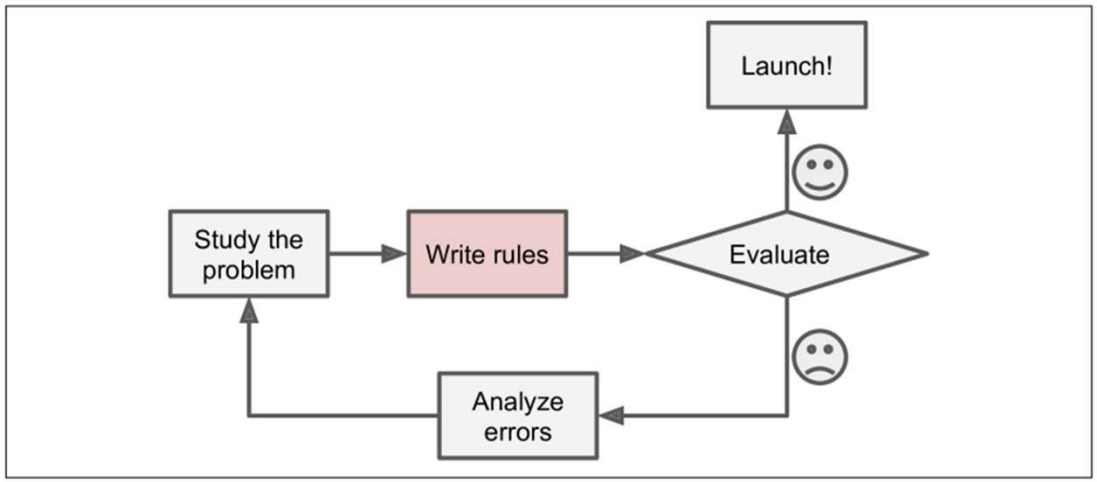
Why Use Machine Learning?

- Rule-based spam filter

  1. Design rules that define spam (e.g. identify words and phrases that are commonly present in spam emails)

  2. Write a decision algorithm for each of the spam patterns that flags email as spam if any of the patterns in (1) are encountered

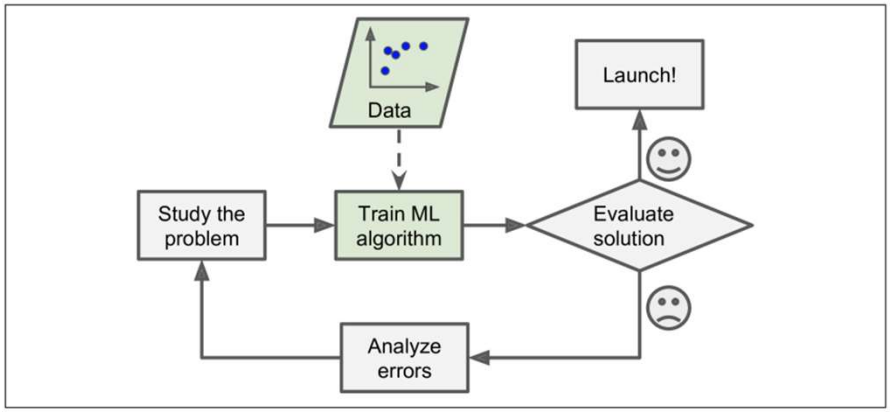
  3. Test the algorithm and repeat steps 1 and 2 until the result is satisfactory enough for production

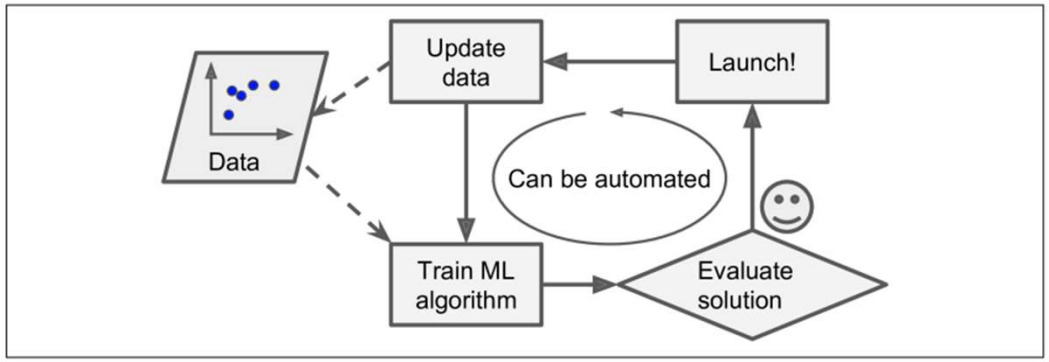


The list of rules most likely will become complex – hard to maintain

Why Use Machine Learning?

- ML-based spam filter



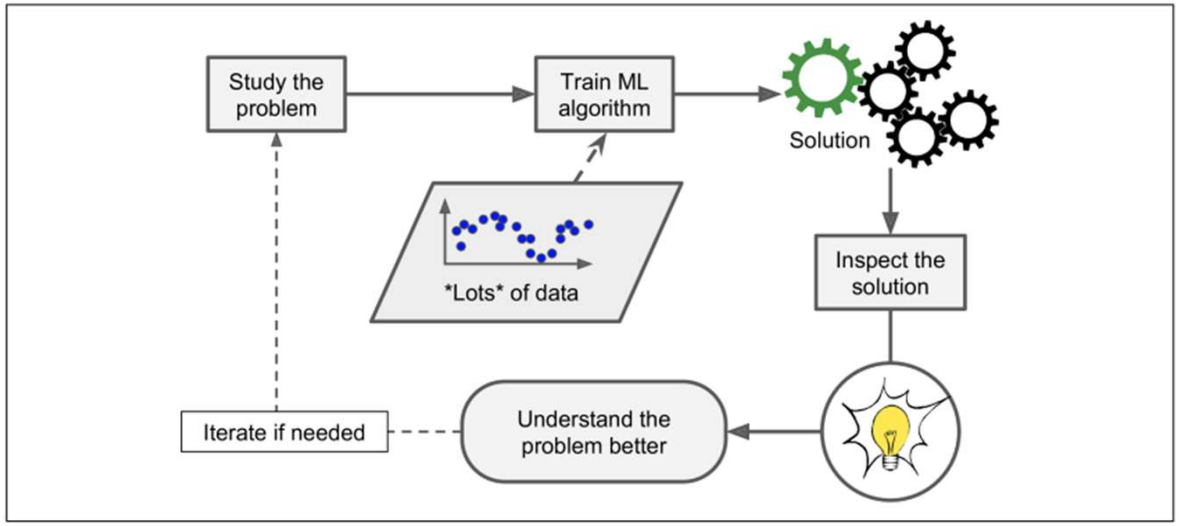


- Obtain relevant data
- Select proper ML algorithm
- Tune the algorithm until satisfied

- New data is handled by re-training the algorithm

Why Use Machine Learning?

- ML and Data Mining



ML algorithm can produce new knowledge

General ML Application Area

- Problems requiring a lot of manual tuning or long lists of rules
- Complex problems with no good solutions based on traditional methods
- Fluctuating environments
- Data mining

Examples of ML Applications

- Image classification on a production line (CNN)
- Analysis of healthcare imagery data (CNN)
- Automatic classification of news articles (NLP)
- Detection of offensive posts in social media (NLP)
- Document summarization (NLP)
- Chatbots (NLU)
- Forecasting of natural, financial, production processes (various regression methods and neural networks)

Examples of ML Applications

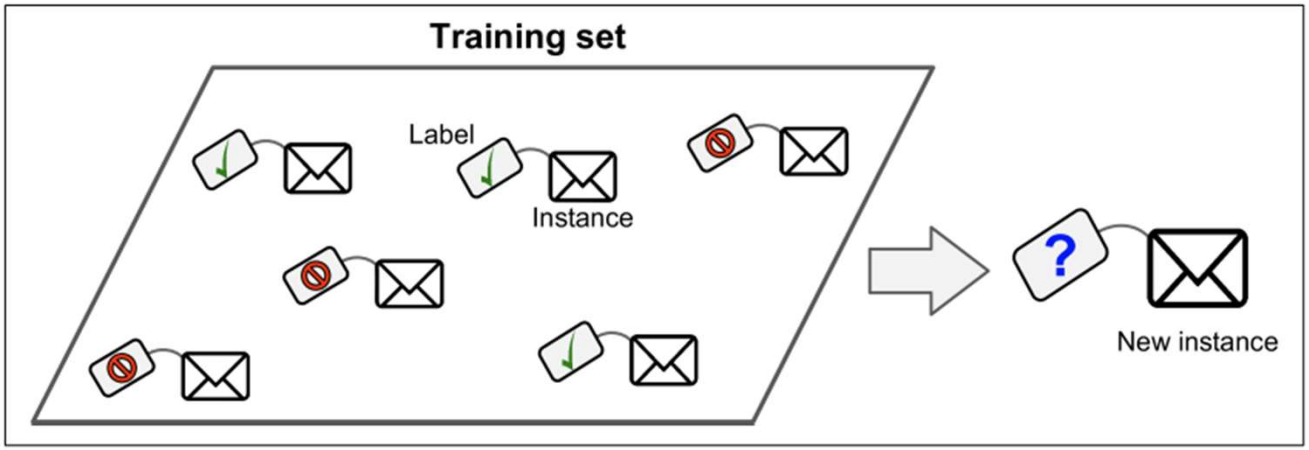
- Voice recognition (RNN, CNN, Transformers)
- Credit card fraud detection (anomaly detection)
- Segmenting clients based on their purchase (clustering)
- Representation of complex high-dimensional data (dimensionality reduction)
- Product recommendation based on past purchases (neural networks)
- Intelligent bots for video games

Types Of Machine Learning Systems

- Supervised vs unsupervised (also, semisupervised, reinforced learning)
- Online vs batch
- Instance-based vs model-based

Supervised ML

- The training set includes the desired solution, called labels



- Typical task: classification

Supervised ML

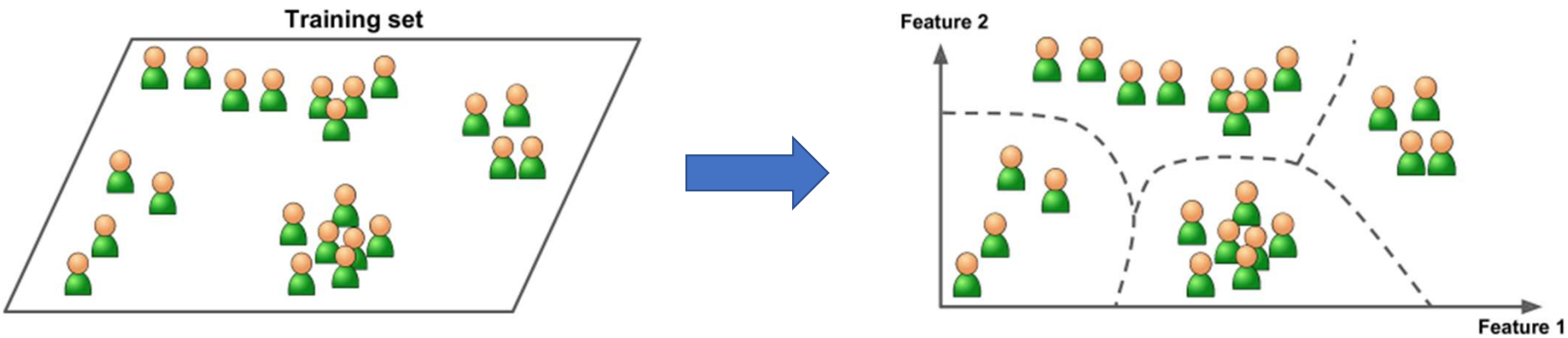
- Can be used to predict numerical values as well

Supervised ML – Examples of Algorithms

- k-Nearest Neighbors
- Linear Regression
- Logistic Regression
- Support Vector Machines (SVMs)
- Decision Trees and Random Forests
- Neural Networks

Unsupervised ML

- The training data is unlabeled

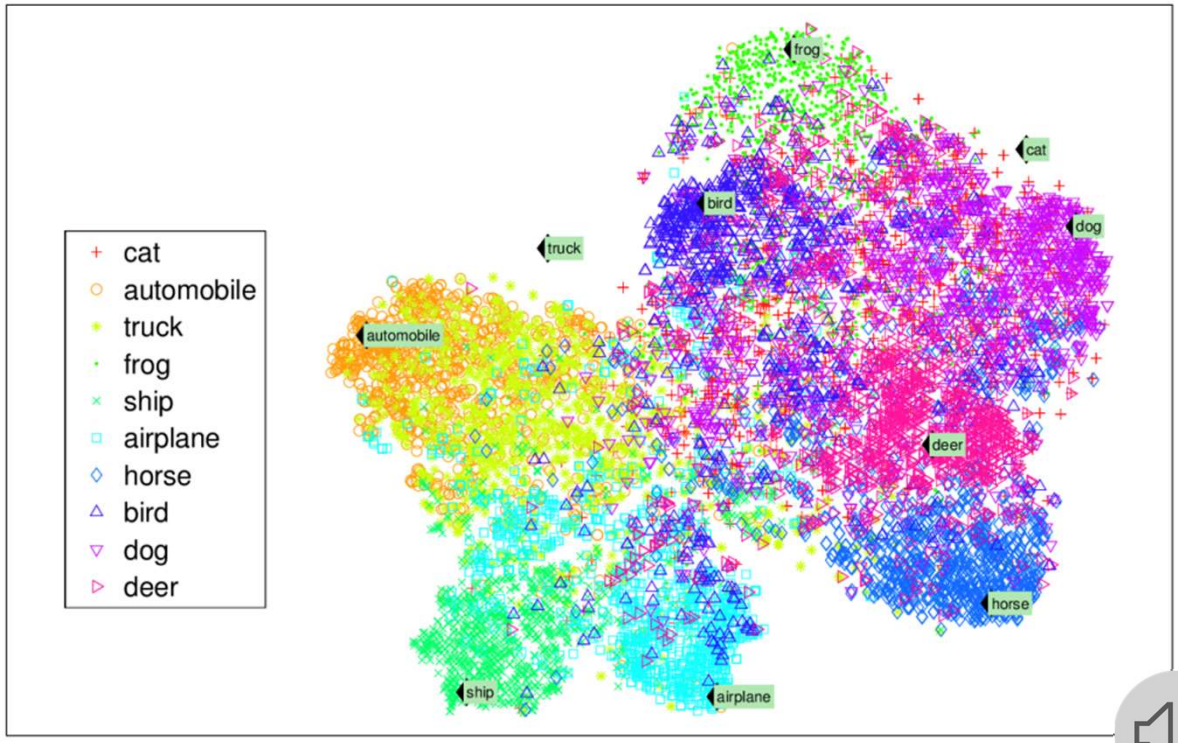


Unsupervised ML – Examples of Algorithms

- Clustering
  - K-Means
  - DBSCAN
  - Hierarchical Cluster Analysis (HCA)
- Anomaly detection and novelty detection
  - One-class SVM
  - Isolation Forest
- Visualization and dimensionality reduction
  - Principal Component Analysis (PCA)
  - Kernel PCA
  - Local Linear Embedding (LLE)
  - t-Distributed Stochastic Neighbor Embedding (t-SNE)
- Association rule learning
  - Apriory
  - Eclat

Unsupervised ML

- Example of a t-SNE (t-distributed Stochastic Neighbor Embedding) visualization highlighting semantic clusters

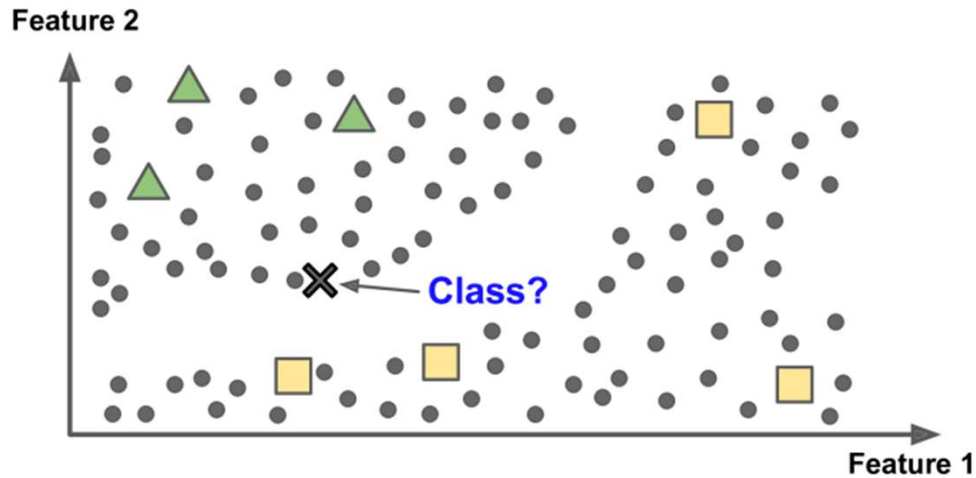


Unsupervised ML

- Anomaly detection
  - Credit card fraud
  - Data outlier detection
  - Quality control
  - Etc.

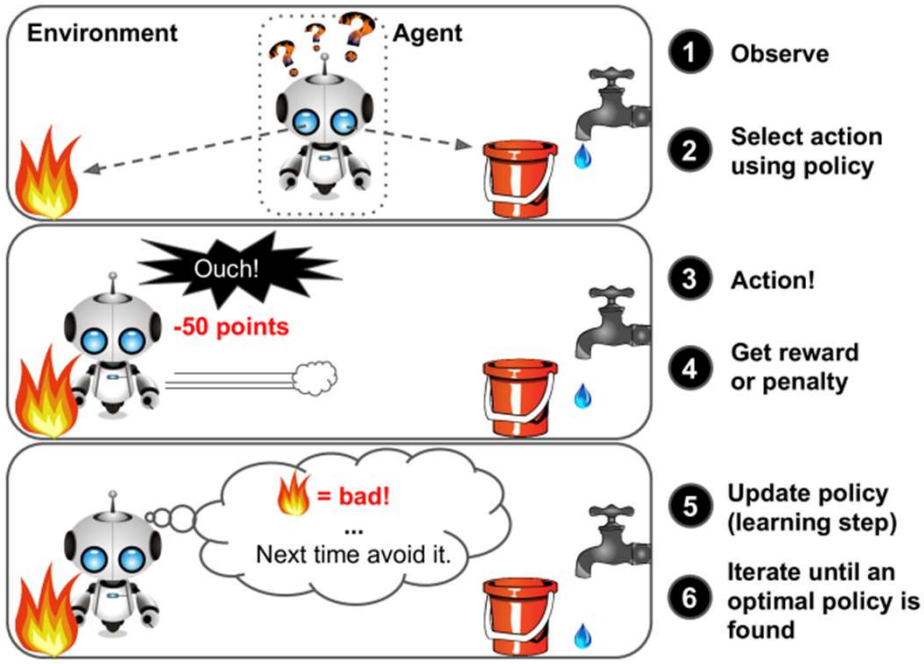
Semisupervised ML

- Partially labeled training data
- Most algorithms are a combination of supervised and unsupervised ML



Reinforcement Learning

- An agent
  - Observes the environment
  - Selects and performs actions
  - Gets reward (or penalty) in return
  - Learns by itself by developing the best strategy (policy) to get the most reward over time

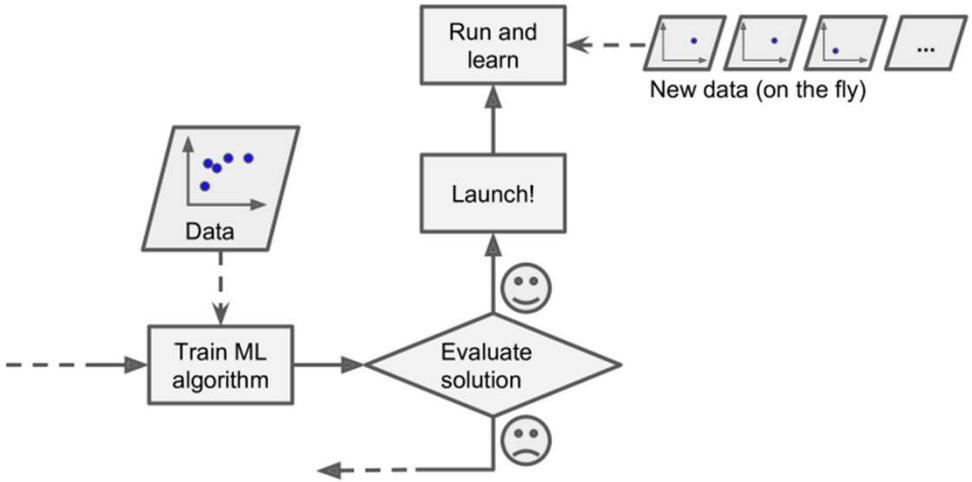


Batch Learning

- To learn, the system must have the entire training dataset
- Takes a lot of time and is usually done offline
- If new data become available:
  - Stop the system
  - Train based on the updated data
  - Re-launch into production

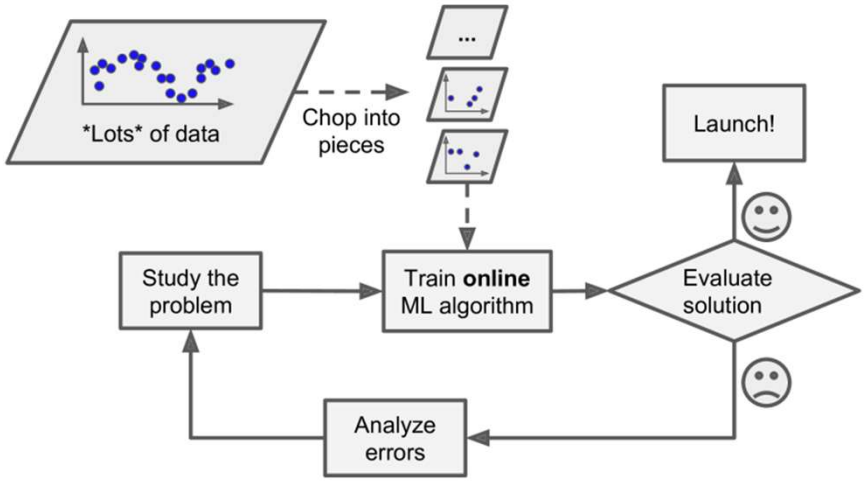
Online Learning

- The system learn incrementally by feeding it data instances sequentially
- Suitable for real-time data applications and platforms with limited resources



Online Learning

- Out-of-core learning
  - Training dataset too large to fit in RAM
  - The dataset is divided into smaller batches and fed to the algorithm
  - Done offline, so a better term is incremental learning



Online Learning

- Learning rate
  - Online learning parameter that defines how fast it adapts to new data (and forgets the old data)
  - In case of bad data injection
    - The system will degrade (speed depends on the learning rate settings)
    - Solution:
      - monitor input data for quality and anomalies
      - monitor the system performance and switch leaning off in case of a degradation

Instance-Based vs Model-Based Learning

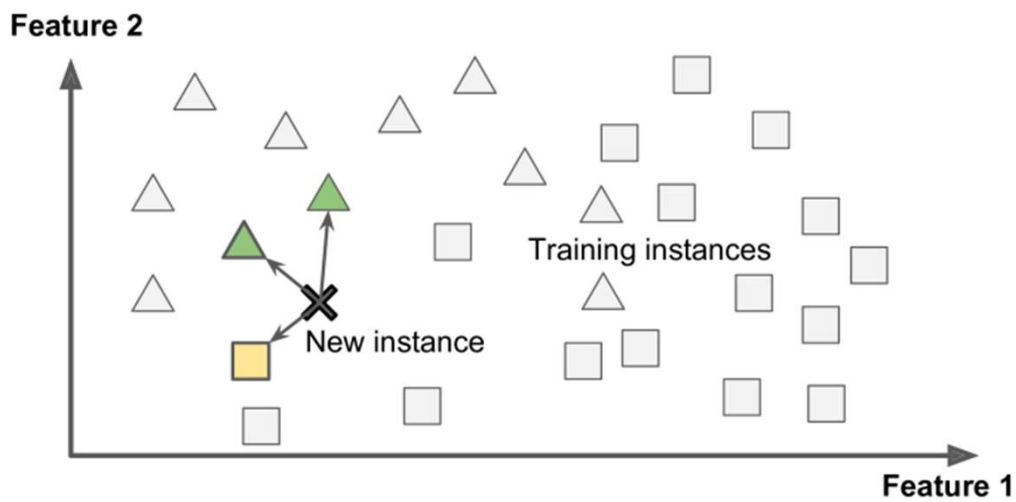
- Main objective of ML is good performance on new data

- Generalization:
  - Ability of a ML system to generalize to examples it has never seen before

- Two main approaches to generalization:
  - Instance-based learning
  - Model-based learning

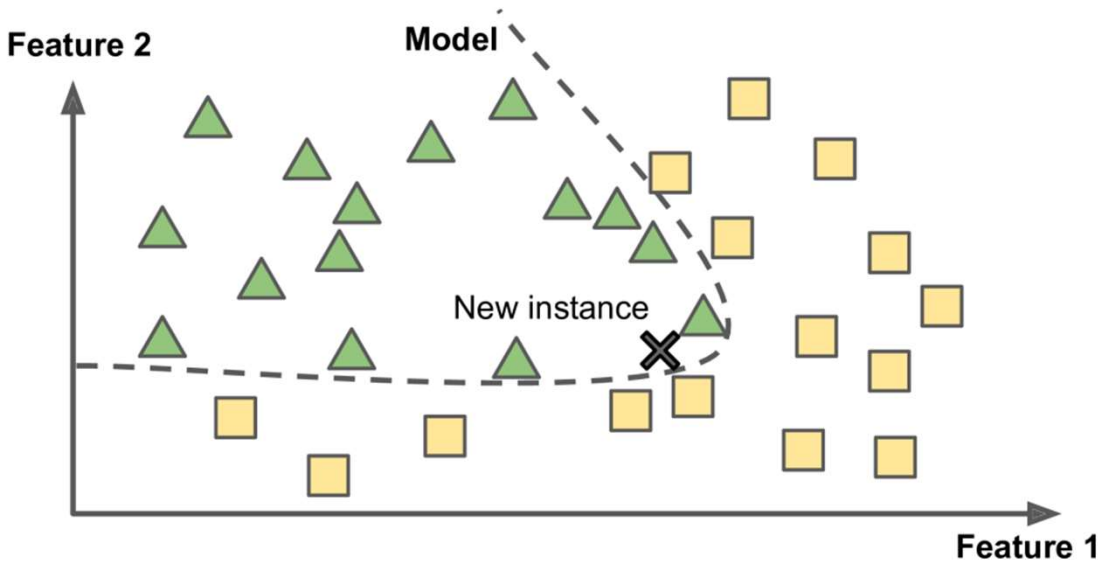
Instance-Based Learning

- Similarity measure
  - New instances are classified by their proximity to the learned examples



Model-Based Learning

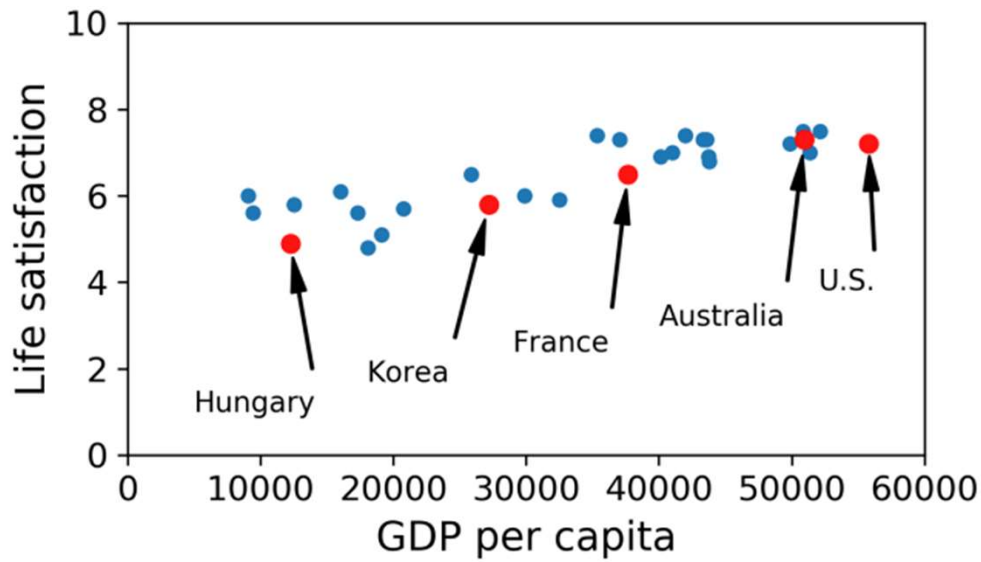
- Build a model based on the training set



Model-Based Learning

- Example: predict life satisfaction based on GDP per capita

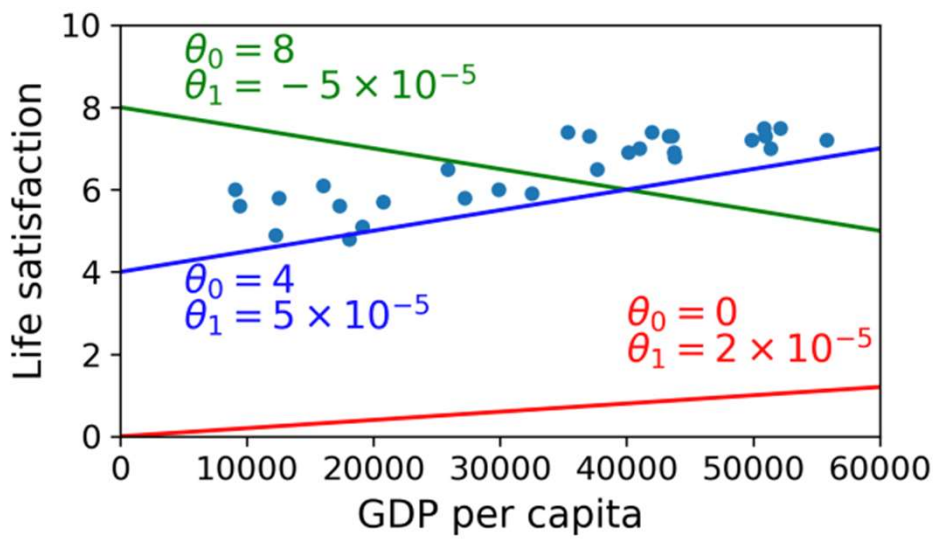
Country	GDP per capita (USD)	Life satisfaction
Hungary	12,240	4.9
Korea	27,195	5.8
France	37,675	6.5
Australia	50,962	7.3
United States	55,805	7.2



Model-Based Learning

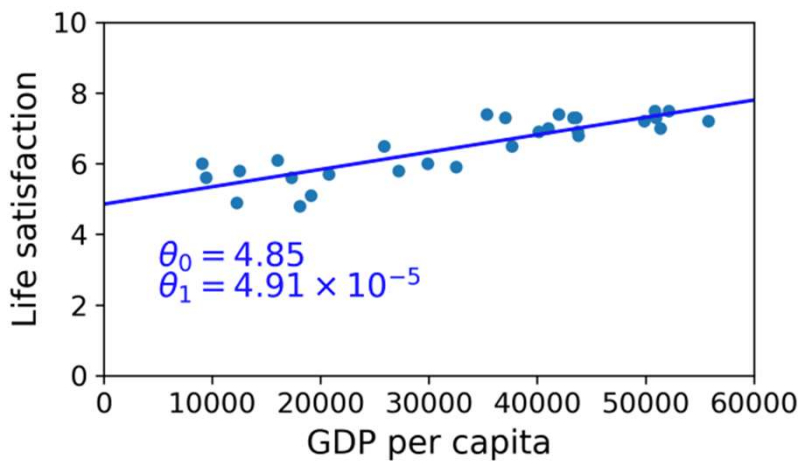
- Example: predict life satisfaction based on GDP per capita
  - Model selection:

$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$



Model-Based Learning

- Example: predict life satisfaction based on GDP per capita
  - Utility (fitness) function:
    - How good the model is at predicting new values
    - Linear regression: an algorithm that selects parameters of the linear function that minimizes distance between predictions and the training examples



Model-Based Learning

- Workflow
  - Study the data
  - Select a model
  - Train the model on the training data
  - Apply the model to new data

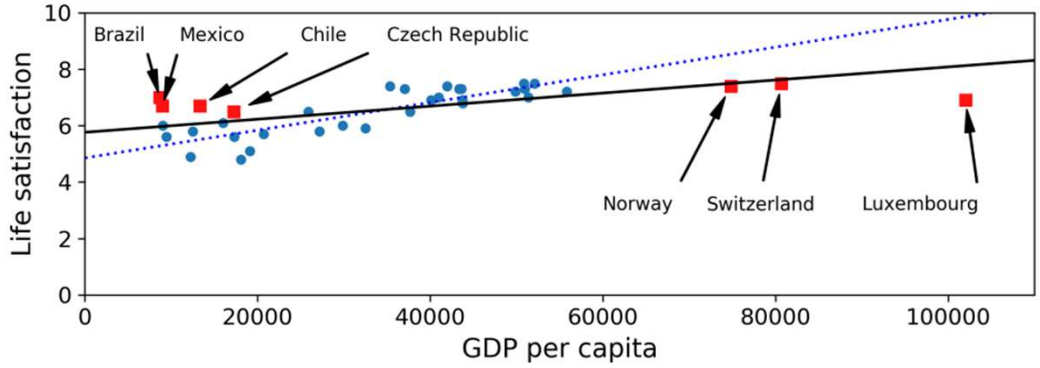
Main Challenges of Machine Learning

Insufficient Quantity of Training Data

- A typical ML algorithm needs thousands or millions of examples to learn properly
- Sufficient corpus volume is not always available
  - Domain of the task is too narrow
  - Events of interest are very uncommon (e.g. failures in manufacturing)
  - Small-scale systems
- Possible solutions
  - Tailor the ML algorithm to the specific task
  - Data augmentation
  - Simulation

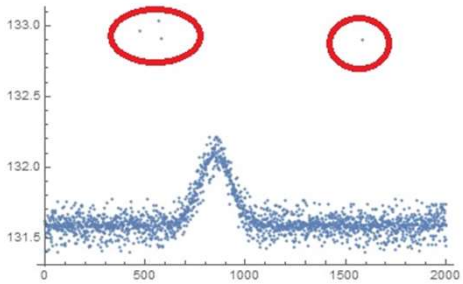
Nonrepresentative Training Data

- Training data must be representative (general enough) of the new cases



Poor-Quality Data

- Outliers



- Errors
  - Missing features
- Noise
  - Low quality measurements

Improving quality of the data can take most of the time spent on solving an ML problem

Irrelevant Features

- Features that are not related to the task
- Features that contain no information

Redundant Features

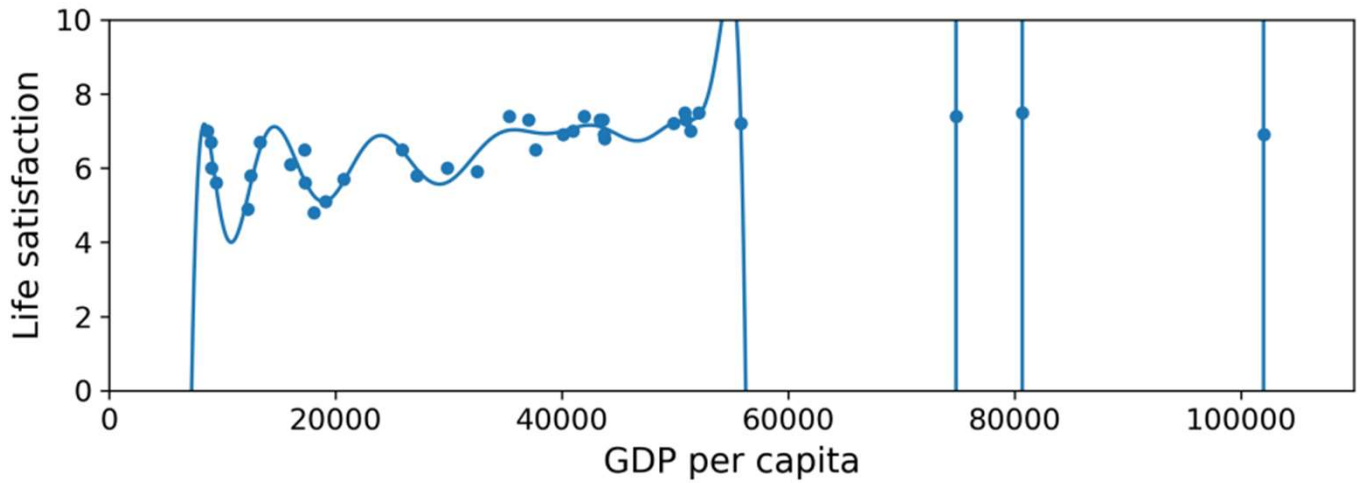
- Features that are strongly correlated with other features in the dataset

Feature Engineering

- Objective:
  - Reduce the number of features (and attributes) to improve ML Model accuracy and reduce complexity of the model
- Steps
  - Feature selection
  - Feature extraction
  - Create new features

Overfitting the Training Data

- The model fits well the training data, but does not generalize well



Overfitting the Training Data

- Complex models can capture subtle patterns in the data
- If the data is of low quality
  - The model can pick up noise instead of the general trend

- In other words
  - Overfitting happens when the model is too complex relative to the amount and noisiness of the training data

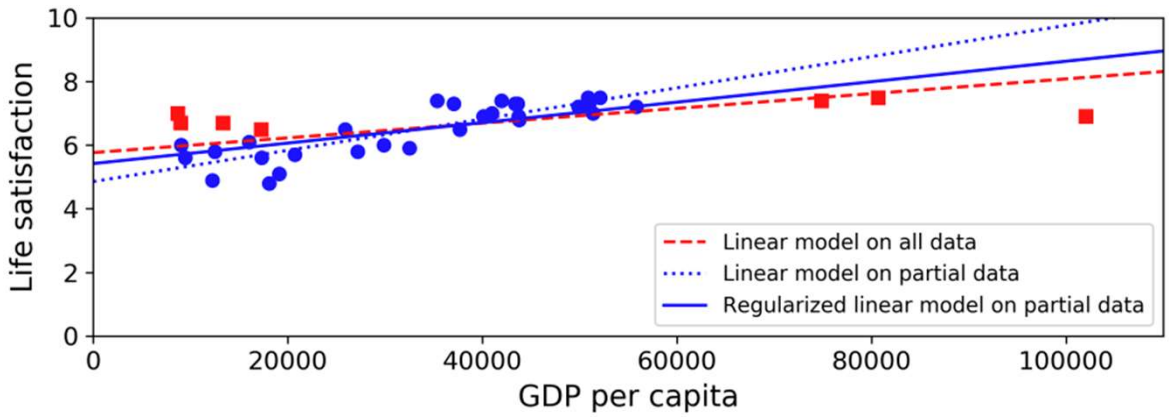
Overfitting the Training Data

- Solution
  - Regularization: simplify the model
  - Gather more data
  - Reduce noise in the training data

Overfitting the Training Data

- Regularization

$$\text{life_satisfaction} = \theta_0 + \theta_1 \times \text{GDP_per_capita}$$



Hyperparameters

- Control the amount of regularization
- Tuning hyperparameters is an important part of building a Machine Learning system

Underfitting the Training Data

- The opposite of overfitting
  - Reality is more complex than the model
- Solution
  - Select more powerful model
  - Feed better features to the algorithm
  - Reduce constraints on the model

Testing and Validating

- Even if a model performs well on the training set, it can fail on new cases – too risky to put in production
- Solution
  - Split the data into two sets
    - Training set
    - Testing set
  - Measure the generalization error
    - Tells how well the model performs on new cases

Testing and Validating – Hyperparameter Tuning and Model Selection

- Performance on the test set indicates only fitness of a particular model with chosen hyperparameters
- Adjusting model choices and hyperparameters on the same test set makes the model overfit for this particular test set
- Solution
  - Holdout validation
    - Set a part of the training set is spared to validate candidate models

Testing and Validating – Hyperparameter Tuning and Model Selection

- Workflow:
  1. Split the training set into a validation set and a reduced training set
  2. Train multiple models with various hyperparameters on the reduced training set
  3. Select the model that performs best on the validation set
  4. Train the best model on the complete training set
  5. Evaluate the model on the test set

Testing and Validating – Hyperparameter Tuning and Model Selection

- Potential problems
  - Validation set too small
  - Validation test too big (and the reduced training set too small)
- Solution: repeated cross-validation
  - Create multiple partitions of the training set into small validation sets and reduced training sets
  - Evaluate each model on every validation set
  - Drawback: training time multiplied by the number of validation sets

Testing and Validating – Data Mismatch

- A data mismatch occurs when the training dataset contains data violating the model application parameters
- Example: a mobile app identifying species of flowers in pictures taken by a smartphone
  - A dataset containing pictures taken by system cameras, drawings of flowers, etc.
- Not always easy to tell
  - The validation test may fail due to either overfitting or the data mismatch

Testing and Validating – Data Mismatch

- Solution
  - Use only representative data for the test and validation sets
  - Hold out part of the training set as a train-dev set
  - Train the model
  - Evaluate on the train-dev set
    - If the model performs well, it is not overfitting
      - Evaluate on the validation set
        - If the model performs poorly – there is data mismatch
    - If the model performs poorly – it is overfitting
      - Simplify, regularize, get more training data, improve training data, etc.